

DATA 266 - HW3

Prompt Engineering and Self-Attention

SID4: 5996 | SEED: 5996

Experiment Report

1. Objective

This assignment investigated two areas of generative AI. First, six prompt-engineering techniques were tested with two examples each using Qwen/Qwen2.5-0.5B-Instruct. Second, single-head scaled dot-product self-attention was implemented from scratch in PyTorch and trained for next-token prediction in both unmasked and causal-masked forms.

2. Experimental Setup

Parameter	Value
SID4 / random seed	5996
Prompt model	Qwen/Qwen2.5-0.5B-Instruct
Execution device	CPU
Text tokens	46
Vocabulary size	39
Next-token input / target length	45 / 45
Embedding dimension	32
Optimizer	Adam
Learning rate	0.01
Loss	CrossEntropyLoss
Training epochs	500

3. Part 1: Prompt Engineering

The six techniques were Zero-Shot, Few-Shot, Chain-of-Thought, Zero-Shot Chain-of-Thought, Meta-Prompting, and Tree of Thoughts. Two tasks were run for each technique, giving 12 prompt experiments.

Technique	Example 1	Example 2	Finding
Zero-Shot	Notebook-cost arithmetic: \$28	Software-update sentiment: Neutral	Direct prompting solved the arithmetic task, but the sentiment output was incorrect; the sentence describes an improvement and would normally be Positive.

Few-Shot	Even/Odd classification for 37	Product-failure sentiment: Negative	The sentiment example followed the demonstrated label pattern. The even/odd example became unnecessarily verbose instead of returning only the expected short label.
Chain-of-Thought	Train distance: 150 miles	Roses/flowers logic: model concluded Yes	Explicit reasoning produced detailed steps, but detail did not guarantee correctness. The roses conclusion does not follow from the premises.
Zero-Shot CoT	17 sheep, all but 9 run away: model answered 8	Age ordering: Carol	The age-ordering task was solved correctly, while the sheep wording was misinterpreted. 'All but 9' means 9 remain.
Meta-Prompting	Ball-count strategy: 10	Mixed camera/battery sentiment: Negative	The model explicitly considered a strategy or relevant clues before answering, producing structured analysis.
Tree of Thoughts	Multiple approaches to 25×16	Three actions for reducing complaints	The model considered multiple alternatives before selecting an approach, but the arithmetic response became repetitive during one branch.

3.1 Prompt-Engineering Findings

- Zero-shot prompting was concise but showed that a direct instruction alone may not be enough for reliable classification.
- Few-shot examples helped communicate the intended task pattern, although the model did not always obey the desired output brevity.
- Chain-of-Thought and Zero-Shot CoT produced visible intermediate reasoning, but the experiments demonstrate that longer reasoning can still contain logical or linguistic errors.
- Meta-prompting encouraged the model to identify a strategy or decision criteria before answering.

- Tree of Thoughts encouraged consideration of alternatives, but additional branching can also make responses longer or repetitive.

4. Part 2: Self-Attention From Scratch

The assignment text was tokenized at the word level, converted to integer token IDs, and used to create 45 next-token input-target pairs. A trainable embedding layer and separate linear Q, K, and V projections were implemented. Attention scores were computed as $QK^T / \sqrt{d}$ and normalized with softmax. No `nn.MultiheadAttention`, `nn.Transformer`, or HuggingFace Transformer attention class was used for this part.

4.1 Unmasked Self-Attention

The unmasked model allowed every query position to attend to every key position. After 500 epochs, the reported cross-entropy loss was 0.3623. The learned attention matrix had shape 45×45 . The notebook heatmap therefore represents learned attention after training rather than random initialization.

Epoch	Unmasked Loss	Causal Loss
50	0.4098	0.0146
100	0.3677	0.0016
150	0.3639	0.0010
200	0.3626	0.0007
250	0.4165	0.0005
300	0.3614	0.0004
350	0.3629	0.0003
400	0.3624	0.0002
450	0.3623	0.0002
500	0.3623	0.0002

4.2 Causal Masked Self-Attention

The causal model applied a lower-triangular mask to the attention-score matrix before softmax. Future-token scores were replaced with negative infinity, so their post-softmax attention weights became zero. After 500 epochs, the reported loss was 0.0002, and the masked attention matrix also had shape 45×45 .

The notebook explicitly verified the masking behavior by measuring the maximum attention assigned to future tokens. The result was 0.0, confirming that no query token attended to a later position.

4.3 Example Learned Causal Attention

For the query token 'they' at the inspected position, attention weights greater than 0.01 were: 'networks' = 0.0443, 'for' = 0.9315, and 'representations' = 0.0185. This provides a concrete example of the learned, non-uniform attention distribution.

5. Comparison and Interpretation

Property	Unmasked	Causal Masked
Can attend to future positions?	Yes	No
Attention matrix	45×45	45×45
Final reported loss	0.3623	0.0002
Future attention restriction	None	Lower-triangular mask before softmax
Maximum future attention	Not restricted	0.0

The two experiments demonstrate the structural effect of causal masking. In the unmasked model, each token can use information from the entire sequence, including later positions. In the causal model, later positions are unavailable by construction. The much lower causal-model training loss is an observed result for this small single-sequence experiment and should not be interpreted as a general claim that causal masking always produces lower loss.

6. Key Findings

- Prompt structure materially changed both the form and quality of model responses.
- More explicit reasoning did not guarantee correctness: the roses logic example and the 'all but 9' sheep example contained reasoning errors.
- Scaled dot-product self-attention was successfully implemented using trainable embeddings and Q, K, and V projections.
- Both attention models produced learned 45×45 attention matrices after training.
- The causal mask behaved correctly: attention to every future position was zero, with maximum measured future attention equal to 0.0.
- The unmasked model ended at loss 0.3623, while the causal model ended at 0.0002 in this experiment.

7. Limitations

The attention experiment uses one short text sequence and a small vocabulary, so the losses and learned patterns should be interpreted only within this controlled assignment. The prompt-engineering results were generated with one small instruction model on CPU and are individual examples rather than a statistical evaluation across many prompts or models.

8. Conclusion

HW3 showed two complementary aspects of generative AI. The prompt experiments demonstrated that prompting technique affects response structure and that additional reasoning can improve transparency without guaranteeing correctness. The attention implementation demonstrated the mechanics of scaled dot-product attention and the role of causal masking in autoregressive prediction. The trained causal model assigned zero attention to future tokens, directly validating the intended mask behavior.

Appendix: Required Artifacts

Repository artifacts: HW_3.ipynb, README.md, RUN_LOG.txt, METRICS.md, AI_USE.md, unmasked_attention_model.pth, and causal_attention_model.pth.